# Chapter 11 — LLM Serving and Inference

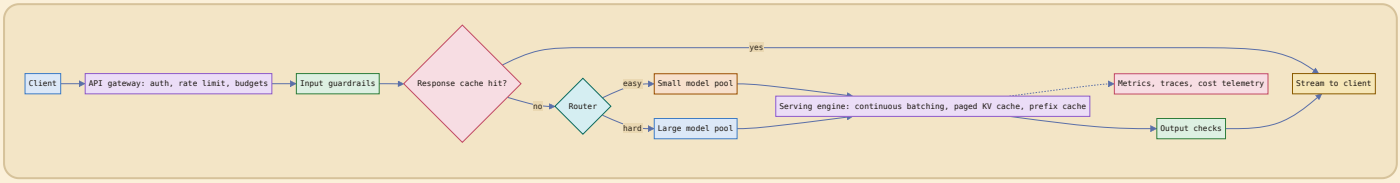
So far an LLM has mostly been a black box you send text to. This chapter opens the box from the **operations** side: what happens on the **GPU** (graphics processing unit — the chip that does the model's maths) when a request arrives, why an answer takes the time it takes, what runs out first when traffic grows, and how teams make generation faster and cheaper without making it worse.

A classic ML service (a fraud score) returns one number in about 10 ms. A **generative** service writes hundreds of **tokens** (word pieces; defined in §11.1), one after another, on expensive **accelerators** (specialised chips such as GPUs), so the same feature can cost a fraction of a cent or several cents per request. **Inference** (running a trained model to get an output) therefore moves to the centre of the design — and good engineers put numbers on it early.

**What you'll learn**

1. **Prefill** vs **decode**, and **latency** metrics (latency = how long a request takes; see System Design Fundamentals) (**TTFT**, **TPOT**, goodput).
2. The **KV cache**, **batching** (static, dynamic, continuous) and **PagedAttention / vLLM**.
3. **Quantization**, **distillation** and **speculative decoding**.
4. **GPU memory sizing** and **model parallelism**.
5. **Streaming**, **caching** (exact, semantic, prefix), **routing and cascades**.
6. **Cost estimation**, **rate limiting** (capping how much each caller may send per minute), API (a provider's hosted model you call over the internet) vs self-hosting (running the model on your own GPUs), **observability** — and a capstone pricing a 5-million-user product.

Prices and throughput figures are **illustrative assumptions** chosen to make the arithmetic easy; learn the relationships and check current prices before relying on them. All code is in `code/`, runs on a laptop **CPU** (ordinary processor) with **NumPy** (Python's numerical-array library), and the outputs shown are real.




The optimizations fall into three groups: skip the expensive call (caching, routing, shorter prompts), use the hardware better (continuous batching, paged KV memory, prefix reuse), and make the model cheaper to run (quantization, distillation, speculative decoding). Start with whichever attacks the largest *measured* cost or latency.

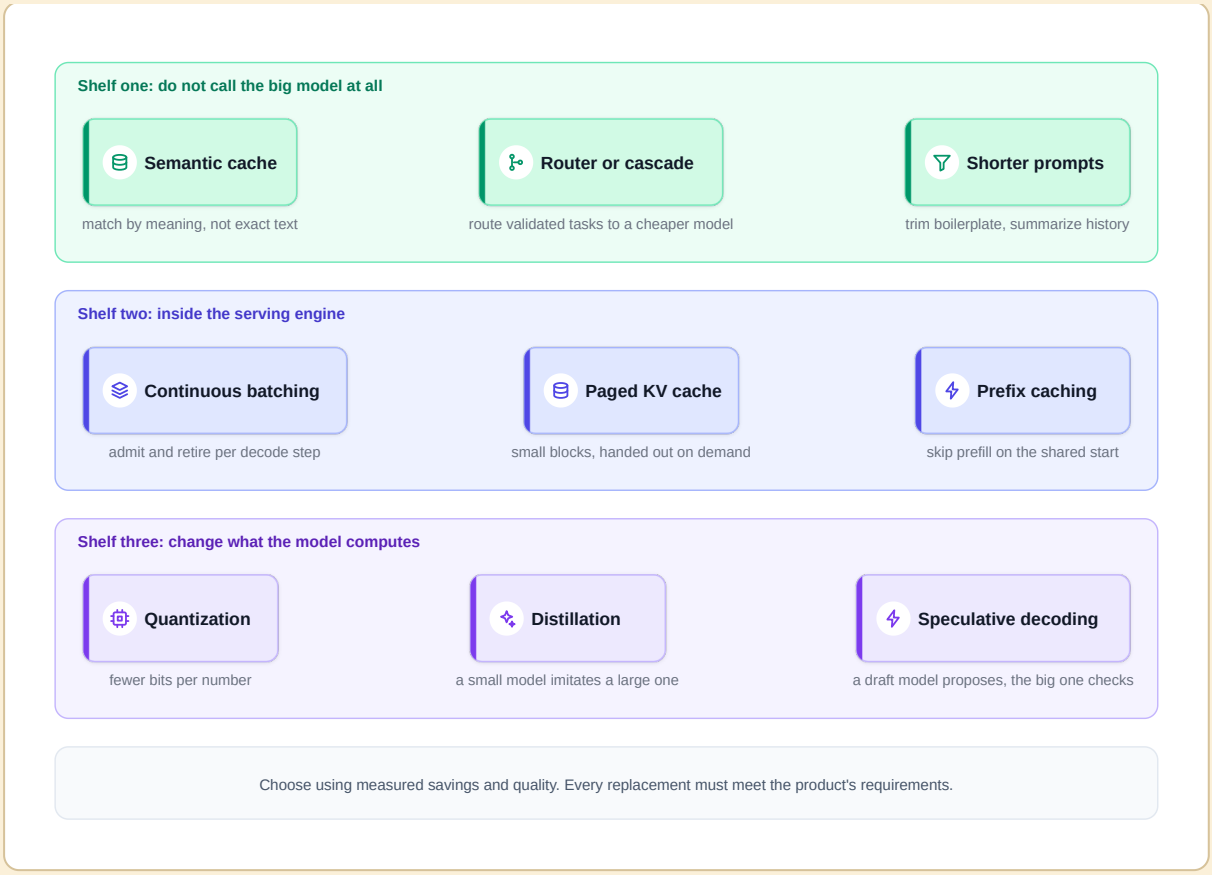



*Figure — The optimization menu, grouped by where each technique works (course diagram).*

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## 11.1 The LLM as a production component

### What it is

To a system designer an LLM **endpoint** (the network address your code sends requests to) is a function: **tokens in → tokens out**, with latency depending on both counts and a price per token. Two terms:

- **Token** — the unit a model reads, writes and bills in. Rough English rule: 1 token ≈ 4 characters ≈ 0.75 words (1,000 words ≈ 1,300 tokens). Code and non-English text use more; use the provider's token counter when accuracy matters.
- **Context window** — the maximum tokens one request may use; input, output and any reasoning tokens usually share it, with a separate cap on output length.

### Why it matters

Everything in this chapter is measured in tokens: latency grows with them, prices are quoted per million of them, the context window caps them. The model does **not** remember earlier conversations — your application must re-send the relevant history each time. Choosing which instructions, history and documents to include (**context assembly**) is one of your biggest cost levers.

### How it works — a request's life

1. The server **tokenizes** the client's text (splits it into tokens and maps each to a number).
2. **Prefill**: the model reads all input tokens and builds internal state (the KV cache, §11.4).
3. **Decode**: it produces output tokens one at a time, each depending on the previous one.
4. Tokens are **streamed** back as produced, or returned together at the end.

5. The bill is `input_tokens × input_price + output_tokens × output_price` (plus any cached-input charges).

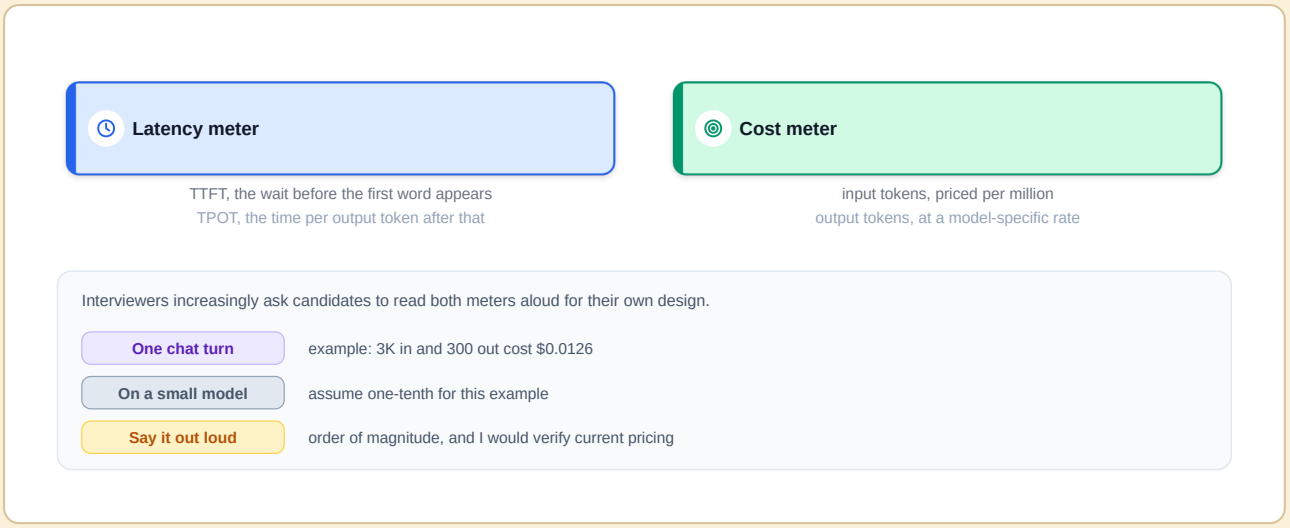



*Figure — Read two meters aloud for any GenAI design: latency and token cost (course diagram).*

### Common mistakes

- Quoting "an LLM call costs X" without the tokens in and out.
- Forgetting that chat history is re-sent every turn (§11.15); treating the context window as memory.

### TL;DR

Latency and cost are both token-shaped. Count tokens first.

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## 11.2 Prefill vs decode

### What it is

Generating a reply happens in two phases with very different hardware behaviour:

| | **Prefill** | **Decode** |
|---|---|---|
| What it does | Processes the whole prompt | Produces one new token per step |
| Parallelism | All prompt positions at once (one big matrix multiply — the GPU's core operation on grids of numbers) | Sequential — token *t+1* needs token *t* |
| Bottleneck | **Compute** (FLOPs — floating-point operations) | **Memory bandwidth** (bytes/s read from GPU memory: weights + KV cache) at small batch |
| Drives | Time to first token | Time per output token |
| Grows with | Input length | Output length |

Analogy: prefill is a chef reading the whole order at a glance; decode is plating one spoonful at a time, walking to the pantry (GPU memory) before each spoonful.

### Why it matters

It explains three facts you will be asked about:

- **The first word takes a moment, then text flows steadily** — the wait is prefill (+ **queueing**, time spent waiting for a free slot); the flow rate is decode.
- **Output tokens cost more than input tokens** — each output token needs its own sequential pass, while input tokens share one parallel pass. (Providers set prices commercially, so the ratio is not physics; list prices typically put output at about 4–8× input.)
- **"Be concise" saves real money** — shorter answers mean fewer decode steps.

## How it works — worked numbers

Take a 7-billion-**parameter** (learned weight) model in **16-bit precision** (each number stored in 16 bits) (2 bytes per parameter → **14 GB of weights**) on a GPU with **~2 TB/s** memory bandwidth (an NVIDIA A100 80 GB; an H100 SXM has ~3.35 TB/s).

1. **Decode, batch of 1.** Each step reads all 14 GB of weights: 14 GB ÷ 2 TB/s = **7 ms per token at best**, ≈ 140 tokens/s for that user. The arithmetic units sit mostly idle — each 2-byte weight feeds only ~2 FLOPs.
2. **Decode, batch of 32.** The same 14 GB read now serves 32 sequences; the step still takes ~7 ms plus extra KV-cache reads, so throughput approaches 32 × 140 ≈ 4,500 tokens/s. This is why **batching** (§11.5) is key.
3. **Prefill of 2,000 tokens.** About 2 × parameters FLOPs per token → $2 \times 7\times10^{9} \times 2{,}000 = 2.8\times10^{13}$ FLOPs. At an effective ~150 **TFLOP/s** (trillion floating-point operations per second) (half the A100's 312 TFLOP/s **BF16** — a 16-bit number format built for ML — peak) that is ≈ 0.19 s, so long prompts show up directly in TTFT.

These are back-of-envelope ceilings, not **benchmarks** (measured runs on real hardware). The toy demo in §11.4 shows the same shape on a CPU: prefill handles ~90,000 prompt tokens/s in one pass, decode only ~5,000 tokens/s one at a time.

## Trade-offs

RAG (retrieval-augmented generation: retrieved documents are pasted into the prompt) and summarization are **prefill-heavy**; chat and code generation with long answers are **decode-heavy** — benchmark both. Engines use **chunked prefill** (split a long prompt into pieces interleaved with other users' decode steps so their output does not stall); some large deployments use **disaggregated serving** (separate GPU pools for prefill and decode).

## In the interview

> **30-second answer:** "Prefill reads the whole prompt in parallel, is compute-bound and sets time to first token. Decode emits one token per step and, at small batch, is limited by memory bandwidth because every step re-reads the weights — it sets streaming speed. That's why output tokens cost more and why batching matters."

- *Why is decode memory-bound?* Each step reads every weight but does only ~2 FLOPs per weight per sequence; the GPU computes far faster than it can stream weights, until the batch is large.
- *How would you cut TTFT for a RAG app?* Fewer/shorter retrieved chunks, prefix caching, chunked prefill, a smaller model or more GPUs per **replica** (one running copy of the model server).

## Self-check

1. Which phase dominates "summarize this 50-page PDF in one sentence"? → Prefill.
2. Why does a batch of 32 barely slow each decode step? → Reading the weights, the expensive part, is shared.

### TL;DR

Prefill = parallel, compute-bound, sets TTFT. Decode = sequential, bandwidth-bound, sets tokens/s. Batch to use the idle compute.

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## 11.3 Latency and throughput metrics

### What it is

| Metric | Meaning | Typical target (chat, illustrative) |
|---|---|---|
| **TTFT** — time to first token | Request sent → first output token arrives (queueing + prefill + network) | < 0.5–1 s |
| **TPOT** — time per output token (≈ **ITL**, inter-token latency) | Average gap between output tokens | 20–50 ms (20–50 tokens/s, faster than reading) |
| **End-to-end latency** | Request sent → last token | Depends on answer length |
| **Tokens/s per user** | 1 / TPOT | — |
| **Throughput** (work done per second) | Total tokens/s (or requests/s) the server produces | Maximize, subject to the above |
| **Goodput** | Requests/s that **meet** their latency targets | What actually matters |

A handy formula: **E2E ≈ TTFT + TPOT × (output_tokens − 1)**. Example: TTFT 0.4 s, TPOT 30 ms, 300 tokens → 0.4 + 0.03 × 299 ≈ **9.4 s**. The user starts reading after 0.4 s — why streaming makes it feel fast.

### Why it matters

Per-user speed and total throughput pull in opposite directions: bigger batches raise throughput but slow each step a little. Report **percentiles**, not averages — p95 is the value 95% of requests beat, and a p99 TTFT of 8 s is what users complain about. An **SLO** (service level objective) such as "p95 TTFT < 1 s and p95 TPOT < 50 ms" makes this a measurable contract; **goodput** counts only requests that meet it.

### Common mistakes

- One "latency" number for a streaming product; benchmarking one request at a time (misses queueing) or with unrealistic lengths.

### Self-check

1. TTFT 1 s, TPOT 25 ms, 500 output tokens — E2E? → 1 + 0.025 × 499 ≈ 13.5 s.
2. Why goodput over throughput? → Throughput also counts late answers.

### TL;DR

Measure TTFT, TPOT and E2E percentiles under realistic load; optimize goodput.

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## 11.4 The KV cache

### What it is

In a transformer (Chapter 3), each token's **attention** step (how it gathers information from other tokens) compares its **query** vector against the **key** vectors of all earlier tokens and mixes their **value** vectors. When generating token 101, the keys and values of tokens 1–100 are unchanged — so the engine **stores** them instead of recomputing. That store, per layer, is the **KV cache**.

Analogy: meeting minutes. Without notes you replay the whole recording before saying anything new; with notes you read them and add one line.

### The problem it solves

Without a cache, token $t$ means pushing all $t$ tokens through the model again, so $n$ tokens cost $1 + 2 + \dots + n \approx n^2/2$ token-passes. With a cache each new token costs one pass (plus reading the cache): linear in $n$. The price is **memory**.

### How it works

1. **Prefill** computes K and V for every prompt position in every layer and stores them.
2. Each **decode** step computes Q, K, V for the **one** new token, appends K and V, and attends over the whole cache.
3. When the request ends, its cache is freed (or kept for prefix caching, §11.13).

**Memory formula (per token):**

```
KV bytes per token = 2 (K and V) × n_layers × n_kv_heads × head_dim × bytes_per_value
KV bytes per request = KV bytes per token × tokens in the sequence (prompt + output so far)
```

Worked examples at 16-bit (2 bytes), using published model shapes:

| Model | Layers | KV heads × head dim | Per token | One 4,096-token sequence |
|---|---|---|---|---|
| Llama-2-7B (standard multi-head attention: many attention units, or heads, in parallel) | 32 | 32 × 128 | 2·32·32·128·2 = 524,288 B = **512 KiB** | **2.0 GiB** |
| Llama-3-8B (grouped-query attention) | 32 | 8 × 128 | **128 KiB** | **0.5 GiB** |
| Llama-3-70B (grouped-query attention) | 80 | 8 × 128 | **320 KiB** | **1.25 GiB** |

Two lessons. A single long conversation can need gigabytes (KiB = 1,024 bytes; GiB = $1{,}024^3$ bytes). And **grouped-query attention (GQA)** — several query heads sharing one K/V head — cuts Llama-3-8B's cache to a quarter of Llama-2-7B's at the same depth, which is why modern models use GQA (or **multi-query attention**, MQA, with one shared K/V head). Storing the cache in 8 bits (**KV-cache quantization**) halves it again at a small accuracy cost.

### Code — cache vs no cache on a toy transformer

`code/kv_cache_demo.py` builds a 4-layer **decoder** (a generate-left-to-right transformer, like GPT) with random weights and generates 256 tokens both ways.

```python
def block(x, w, k_past=None, v_past=None):
    """One transformer block. x: (T_new, D). Returns output and full K, V."""
    h = rms_norm(x)
    q, k, v = h @ w["q"], h @ w["k"], h @ w["v"]
    if k_past is not None:                      # decode: reuse cached K, V
        k = np.concatenate([k_past, k])
        v = np.concatenate([v_past, v])
    t_new, t_all = q.shape[0], k.shape[0]
    scores = q @ k.T / np.sqrt(D)               # (T_new, T_all)
    mask = np.triu(np.ones((t_new, t_all)), k=t_all - t_new + 1).astype(bool)
    scores[mask] = -1e9                         # causal: no peeking ahead
    x = x + softmax(scores) @ v @ w["o"]
    x = x + np.maximum(0, rms_norm(x) @ w["m1"]) @ w["m2"]
    return x, k, v

def generate_no_cache(prompt, n_new):          # re-run the whole sequence each step
    seq = list(prompt)
    for _ in range(n_new):
        x = emb[seq] + pos_enc(np.arange(len(seq)))
        for w in W:
            x, _, _ = block(x, w)
        seq.append(int(np.argmax(rms_norm(x[-1]) @ out_head)))
    return seq[len(prompt):]

def generate_with_cache(prompt, n_new):
    x = emb[prompt] + pos_enc(np.arange(len(prompt)))     # prefill: one parallel pass
    cache = []
    for w in W:
        x, k, v = block(x, w)
        cache.append((k, v))
    out = [int(np.argmax(rms_norm(x[-1]) @ out_head))]
    for step in range(n_new - 1):                          # decode: one token per step
        x = emb[[out[-1]]] + pos_enc(np.array([len(prompt) + step]))
        for i, w in enumerate(W):
            x, k, v = block(x, w, *cache[i])
            cache[i] = (k, v)
        out.append(int(np.argmax(rms_norm(x[-1]) @ out_head)))
    # ...
```

What each part does: `block` optionally receives the past K and V and appends the new token's K/V; the causal mask lets each position see only itself and earlier positions. The no-cache generator recomputes everything each step; the cached one does one prefill, then pushes one token per step.

Output:

```
prompt tokens=32  generated=256  layers=4  d_model=128
identical output tokens: True
no cache :   0.53 s  (  481.9 tokens/s)
KV cache :   0.05 s  ( 5041.0 tokens/s)
speed-up : 10.5x
KV cache holds 293,888 floats = 2 (K,V) x 4 layers x 287 tokens x 128
  = 574 KiB at fp16
token positions pushed through the model: no cache=40,832  cache=287
prefill: 32 tokens in 0.34 ms -> 94,263 tokens/s (one parallel pass)
decode : 255 tokens one by one -> 5,021 tokens/s
```

Same tokens, ~10× faster, because the model processed 287 token positions instead of 40,832 (timings vary a little per run). The gap grows with sequence length, so no real engine serves without a KV cache.

### Trade-offs

Compute drops from ~$n^2/2$ to ~$n$ token passes; memory grows with users × context, and cache memory (not compute) becomes the concurrency limit.

### Common mistakes

- Sizing GPUs for weights only; forgetting that a 128K-token context needs 32× the cache of a 4K one.
- Confusing the **KV cache** (inside the engine) with a **response cache** (in front of the model, §11.13).

### Real-world examples

Every production engine (vLLM, TensorRT-LLM, SGLang, llama.cpp) keeps a KV cache; **Hugging Face** (the main hub and Python library for open models) `transformers` calls it `past_key_values` and uses it by default in `generate()`. GQA is used by Llama 2 70B, the Llama 3 family and Mistral 7B.

### In the interview

> **30-second answer:** "The KV cache stores each layer's keys and values for tokens already processed, so each decode step computes them only for the newest token — quadratic recomputation becomes linear work. It costs 2 × layers × KV-heads × head-dim × bytes per token, about 320 KiB per token for Llama-3-70B, so it, not the weights, usually limits concurrent users per GPU."

- *How do you shrink it?* GQA/MQA, 8-bit KV cache, shorter contexts, paged allocation (§11.6), **offloading** (moving) old blocks to CPU memory.
- *Why does it limit batch size?* Each sequence needs its own cache; when memory runs out the engine queues or **pre-empts** requests (pauses one and evicts its cache to make room).

### Self-check

1. KV per token for a model with 40 layers, 8 KV heads, head dim 128, fp16? → 2·40·8·128·2 = 163,840 B = 160 KiB.
2. Does the KV cache change the output? → No — identical tokens, as the demo shows.
3. What does GQA trade away? → A little modelling capacity for a much smaller cache.

### TL;DR

Compute K and V once, reuse them every step. 2·L·H_kv·d·bytes per token, times users × context — usually the real capacity limit.

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## 11.5 Batching: static, dynamic and continuous

### What it is

**Batching** runs several requests through one **forward pass** (one run of inputs through all the model's layers) so one read of the weights serves many sequences (§11.2).

- **Static batching** — collect a fixed-size batch, run it until **every** sequence finishes, then take the next batch.
- **Dynamic batching** — start when the batch is full **or** the oldest request has waited a maximum time (e.g. 200 ms); still runs to completion.
- **Continuous batching** (**in-flight** or **iteration-level** batching, from the Orca system, OSDI 2022) — the **scheduler** (the engine component that decides which sequences run in each step) adds and removes sequences **at every decode step**, so a waiting request takes a freed slot immediately.

Analogy: a tour bus that returns only when the last tourist is back (static) versus a hop-on/hop-off bus (continuous).

## The problem it solves

Output lengths vary wildly ("yes" vs a 600-token essay). With static or dynamic batching, finished sequences hold idle slots until the longest ends, and new arrivals wait for the whole batch. Continuous batching removes both wastes.




*Figure — Continuous batching admits and retires requests every token (course diagram).*

### Code — a small simulator

`code/batching_sim.py` models one GPU with 16 slots. A decode step gives every sequence one token and costs `25 ms + 0.5 ms × batch size` (nearly flat, since decode is memory-bound). 400 requests arrive at random, each wanting 20–500 tokens.

```
if policy == "continuous":                # refill any free slot, every step
    while queue and len(running) < SLOTS:
        r = queue.pop(0)
        running[r] = lengths[r]
elif not running:                         # static/dynamic: only between batches
    # static waits for 16 requests; dynamic starts after 16 requests OR 200 ms
    ...
# one decode step: every active sequence gets one token
# static/dynamic keep finished sequences' slots until the whole batch ends
```

Output:

```
400 requests, 20-500 output tokens each (mean 256), 16 slots per GPU
step time = 25.0 ms + 0.5 ms per sequence in the batch

--- load: 0.8 req/s = 205 output tok/s demanded ---
metric                  static     dynamic  continuous
throughput tok/s           194         192         196
slot utilisation           55%         36%         34%
mean TTFT s              10.10        7.45        0.04
p95 TTFT s               21.77       13.79        0.05
mean latency s           18.53       15.29        7.26

--- load: 1.6 req/s = 410 output tok/s demanded ---
metric                  static     dynamic  continuous
throughput tok/s           260         259         382
slot utilisation           55%         53%         74%
mean TTFT s              65.60       66.68        0.30
p95 TTFT s              123.36      124.45        1.95
mean latency s           74.03       75.10        8.32

batch of 1 would give 39 tok/s; a full batch of 16 gives 485 tok/s
```

How to read it:

- **Light load:** all three keep up with demand, but static batching makes people wait ~10 s for a first token (it waits for 16 requests); dynamic's 200 ms timer helps but new requests still wait behind long batches; continuous starts in ~40 ms.
- **Heavy load:** static and dynamic cap out near 260 tokens/s because ~45% of slot-steps are wasted on finished sequences, and TTFT passes a minute. Continuous delivers ~380 tokens/s at 74% slot utilisation with p95 TTFT under 2 s.
- **Batching at all:** one sequence at a time gives 39 tokens/s; a full batch gives 485 — 12× from the same hardware.

## Trade-offs

| | Static | Dynamic | Continuous |
|---|---|---|---|
| Good for | Offline jobs, similar lengths | Fixed-cost models (classifiers, embeddings) | LLM serving — the default today |
| Weakness | Idle slots, long waits | Wastes slots on finished sequences | Needs per-step scheduling; per-user speed dips as the batch grows |

## When not to use it

For **embedding** or **classification** models (one forward pass, no decode loop), dynamic batching is simpler and sufficient. For **offline** jobs, maximize batch size and ignore TTFT; hosted providers also sell **asynchronous batch APIs** (submit many jobs now, collect results later) at a discount (Anthropic's and OpenAI's are 50% of the standard price) for work that can wait hours.

## Common mistakes

- Comparing engines at batch size 1.
- Unlimited batch growth: throughput rises but TPOT breaks the SLO. Engines cap concurrent sequences (vLLM's `max_num_seqs`) and tokens per step.

## In the interview

**30-second answer:** "Decode underuses the GPU, so we batch. Static batching holds finished slots until the longest sequence ends. Continuous batching schedules per decode step — a slot is

refilled the moment a sequence finishes — raising throughput and cutting queueing when output lengths vary. It's the default in vLLM, SGLang and TensorRT-LLM (which calls it in-flight batching)."

- *Its cost?* Per-user tokens/s drops slightly as the batch grows, and long prefills can stall others' decode — fixed with chunked prefill and batch caps.

### Self-check

1. Why does static batching waste capacity? → Finished answers keep their slots until the longest ends.
2. What does dynamic batching add? → A maximum wait, so small batches can start.

### TL;DR

Batch to share weight reads; continuous batching is the LLM standard.

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## 11.6 PagedAttention, vLLM and serving engines

### What it is

**PagedAttention** (Kwon et al., SOSP 2023 — the paper behind **vLLM**) manages the KV cache the way an **operating system** manages **RAM** (the computer's working memory): in small fixed-size **blocks** (16 tokens by default in vLLM) handed out on demand, with a per-sequence **block table** mapping token positions to physical blocks that need not be **contiguous** (side by side in memory).

### The problem it solves

Earlier engines reserved one contiguous slab per request sized for the **maximum** possible output — say 2,048 tokens — even if the answer ended after 50. The vLLM paper measured that such systems used only about 20–38% of KV memory for real token data, i.e. roughly **60–80% was wasted** by over-reservation and **fragmentation** (free memory broken into gaps too small to use); paging cut the waste to under 4%, fitting more sequences per GPU and giving **2–4× higher throughput** than FasterTransformer and Orca at the same latency.

### How it works

1. Free GPU memory (after weights) is carved into equal KV blocks.
2. A sequence gets blocks only as its tokens arrive; only its last block is partly filled.
3. A finished sequence's blocks return to the free pool.
4. Sequences with the same prefix (or parallel samples of one prompt) **share** physical blocks, copying only when they diverge (**copy-on-write**).
5. If memory runs out, the scheduler **pre-empts** a sequence (swaps its blocks to CPU or recomputes them later).

Paging also makes **prefix caching** cheap: a common system prompt's blocks stay resident and later requests skip that prefill (§11.13). SGLang's **RadixAttention** keeps cached prefixes in a tree for the same purpose.

### Serving engines at a glance

| Engine | Notes |
|---|---|
| **vLLM** (open source: code anyone may use and change) | PagedAttention, continuous batching, prefix caching, many quantization formats, tensor/pipeline parallel, OpenAI-compatible server (accepts the same request format as OpenAI's API) |
| **TensorRT-LLM** (NVIDIA) | Compiled, highly optimized **kernels** (small low-level programs that run one operation on the GPU) for NVIDIA GPUs; "in-flight batching"; often served via Triton |
| **SGLang** (open source) | RadixAttention prefix sharing, fast structured generation |
| **Hugging Face TGI** | Early continuous-batching server; in maintenance mode since December 2025 — Hugging Face now points users to vLLM and SGLang |
| **llama.cpp / Ollama** | Runs GGUF-quantized (llama.cpp's compact model-file format) models on laptops, CPUs and consumer GPUs — the usual choice for local experiments |

Your job is usually to **choose and configure** one of these, not to write kernels. One kernel-level name you will hear: **FlashAttention**, an exact attention algorithm that **tiles** the computation (works on small blocks that fit in fast on-chip memory) to cut slow memory traffic.

### Trade-offs and common mistakes

Paging adds bookkeeping but the memory win dominates. Mistake: assuming paging creates memory — it only stops wasting it.

### In the interview

> **30-second answer:** "PagedAttention stores the KV cache in small blocks allocated on demand instead of a max-length slab per request. That removes most fragmentation, lets requests share prefix blocks, and fits far more concurrent sequences per GPU — the core of vLLM."

### Self-check

1. What OS idea does PagedAttention borrow? → Virtual-memory paging with a page (block) table.
2. How does it help prefix caching? → Many sequences can point at the same prefix blocks.

### TL;DR

KV memory in pages, not slabs: less waste, more users, shared prefixes.

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## 11.7 Quantization

### What it is

**Quantization** stores numbers with fewer bits. Weights are normally 16-bit **floats** (numbers with a fractional part; FP16 or BF16); quantized weights are 8- or 4-bit **integers** (whole numbers) plus a **scale** factor mapping them back to real values. Analogy: recording heights to the nearest centimetre instead of micrometre — far less to write, almost always good enough.

### Why it matters

- **Memory:** raw weights shrink to ½ (int8) or ¼ (int4) of 16-bit: a 70B model goes from ~140 GB to ~70 GB or ~35 GB.

- **Speed:** decode is bandwidth-bound, so reading half the bytes per step can make it faster — *if* the hardware and kernels support the format.
- **Room for KV cache:** a 70B model on 2×80 GB fits only 2 concurrent 4K-token sequences at fp16 but 54 at int8 (§11.10).

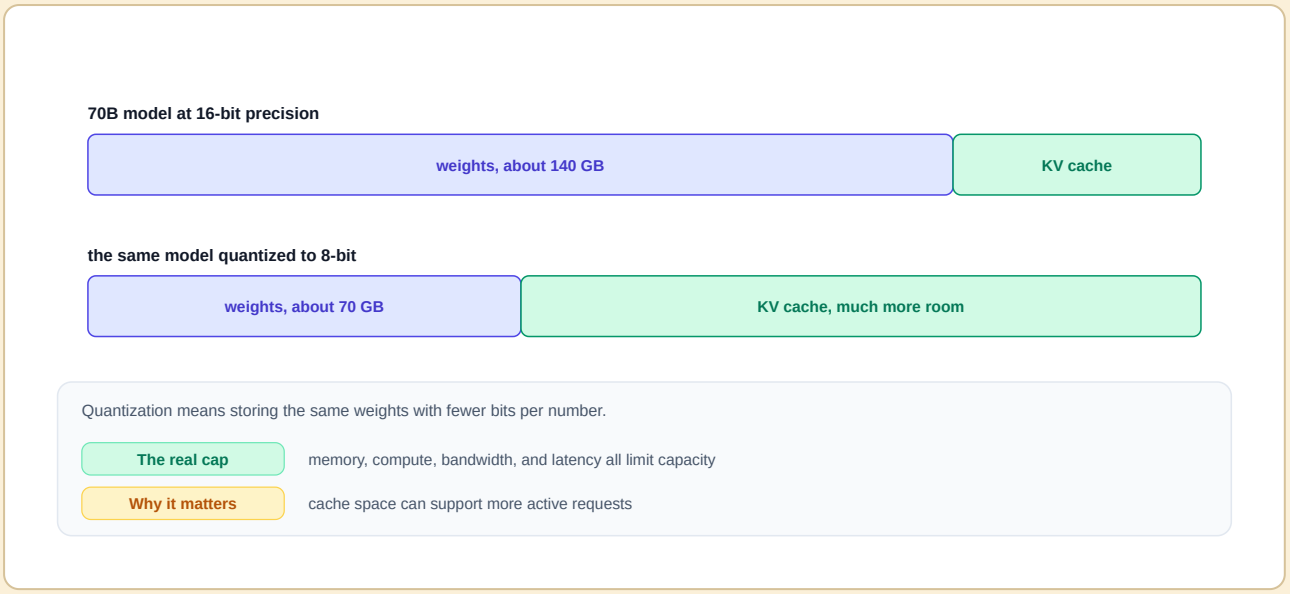



*Figure — A 70B model's raw weights at 16-bit vs 8-bit (course diagram).*

## How it works — symmetric "absmax" quantization

1. Pick a group of weights (the whole **tensor** — a multi-dimensional array of numbers — one output channel, or a small group such as 128 weights).
2. `scale = max(|w|) / q_max`, where `q_max = 127` for int8 and `7` for int4.
3. Store `q = round(w / scale)` as a small integer.
4. At run time use `ŵ = q × scale`.

Worked example (int4, one group): weights `[0.12, −0.03, 0.40, −0.21]`. max|w| = 0.40 → scale = 0.40/7 ≈ 0.0571. q = round(w/scale) = `[2, −1, 7, −4]`. Dequantized: `[0.114, −0.057, 0.400, −0.229]`. Errors are at most half a step (≈ 0.029).

The catch is **outliers**: one huge weight makes the scale huge, so the small weights round to zero. Smaller groups (one scale per 32–128 weights) contain the damage at the cost of more scales.

### Code — measuring quantization error

`code/quantization_demo.py` builds a 1024×1024 weight matrix (normal, σ = 0.02) with 20 planted outliers, quantizes it several ways, and measures the relative error of the weights and of the output `y = x @ W`.

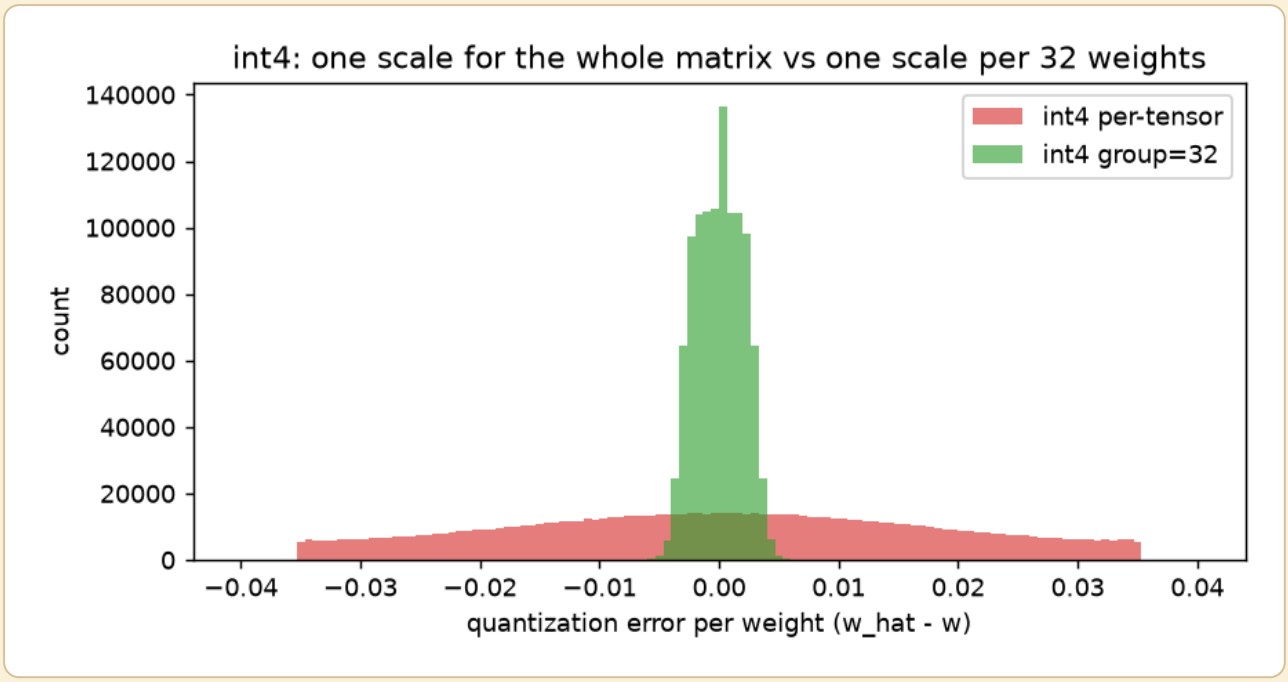
```
def quantize(w, bits, group=None, axis_channel=False):
    q_max = 2 ** (bits - 1) - 1
    if axis_channel:                                    # one scale per output column
        scale = np.abs(w).max(axis=0, keepdims=True) / q_max
        return np.clip(np.round(w / scale), -q_max, q_max) * scale
    if group:                                           # one scale per `group` weights
        wg = w.reshape(-1, group, w.shape[1])
        scale = np.abs(wg).max(axis=1, keepdims=True) / q_max
        return (np.clip(np.round(wg / scale), -q_max, q_max) * scale).reshape(w.shape)
    scale = np.abs(w).max() / q_max                     # one scale for everything
    return np.clip(np.round(w / scale), -q_max, q_max) * scale
```

Output:

```
scheme             bytes/param  weight err  output err
fp16 (baseline)          2.000      0.0000      0.0000
int8 per-tensor          1.000      0.0559      0.0557
int8 per-channel         1.000      0.0100      0.0101
int4 per-tensor          0.500      0.8790      0.8814
int4 per-channel         0.500      0.1767      0.1768
int4 group=128           0.516      0.1227      0.1223
int4 group=32            0.562      0.0985      0.0985

per-tensor int4 scale = 0.0705; typical |w| = 0.0135
weights rounded to 0 by per-tensor int4: 92.2%

70B model, raw weight memory:
  fp16        140.0 GB
  int8         70.0 GB
  int4 g128    36.1 GB
saved img/quantization_error.png
```




Reading the results: int8 with per-channel scales is almost lossless (1% error). Naive int4 with one scale for the whole matrix is a disaster — the outliers make the step so large that **92% of weights become zero**. Group-wise scales bring int4 error to ~10–12% for a little more storage (a 16-bit scale per 128 weights adds 0.125 bits per weight). That error comes from plain round-to-nearest; the methods below reduce its *effect* on the model's output.

### Smarter methods (intuition only)

| Method | Idea in one line |
|---|---|
| **LLM.int8()** (Dettmers et al., 2022) | Keep the few outlier **activation** channels (activations = the intermediate values flowing between layers) in 16-bit and do the rest in int8. |
| **GPTQ** (Frantar et al., 2022) | Quantize one column at a time and **adjust the not-yet-quantized weights** to cancel the error, using second-order statistics (how sensitive the output is to each weight) from a small **calibration set** (a sample of realistic inputs). One-shot, no retraining; popular for 4-bit. |
| **AWQ** (Lin et al., 2023) | "Activation-aware": the ~0.1–1% of weight channels that meet large activations matter most, so **scale them up before quantizing** (and the input down to compensate). No backprop. |
| **SmoothQuant** (Xiao et al., 2022) | Move activation outliers into the weights with a per-channel rescale so both **weights and activations** can be int8 (W8A8). |
| **FP8** | 8-bit floating point, native on recent GPUs (e.g. H100); little quality loss. |
| **GGUF k-quants** (llama.cpp) | Block-wise 2–8-bit formats for CPU/consumer GPU, e.g. `Q4_K_M`. |

Keep apart **weight-only** quantization (W4A16, W8A16 — saves memory and bandwidth, maths still 16-bit) and **weight + activation** quantization (W8A8, FP8 — also uses faster low-precision maths units).

## Trade-offs

| | fp16/bf16 | int8 / fp8 | int4 (GPTQ/AWQ) |
|---|---|---|---|
| Weight memory | 1× | ½× | ~¼× |
| Quality | Reference | Usually near-identical | Measurable drop, larger on small models and hard tasks (maths, code) |
| Speed | Baseline | Faster where kernels support it | Faster decode at small batch; dequantization can hurt at large batch |

## When not to use it

When memory and bandwidth are not your bottleneck, or your eval shows a drop on a sensitive task. Never ship a quantized model without re-running your task evaluation — **perplexity** (a generic measure of how well the model predicts text) can hide task-specific losses.

## Common mistakes

- Assuming int4 = 4× faster — savings are in memory; speed depends on kernels and batch size.
- Forgetting scales and the KV cache in memory maths; calibrating GPTQ/AWQ on data unlike your traffic.

## Real-world examples

Hugging Face hosts thousands of GPTQ/AWQ/GGUF **checkpoints** (saved model files); vLLM and SGLang load AWQ/GPTQ/FP8 directly; Ollama ships 4-bit GGUF models by default so 7–8B models run on laptops; QLoRA fine-tunes **adapters** (small trainable add-on layers) on a 4-bit base model (Chapter 8).

## In the interview

> **30-second answer:** "Quantization stores weights — optionally activations or the KV cache — in 8 or 4 bits with per-group scales. It halves or quarters weight memory and, since decode is

bandwidth-bound, often speeds it up. The risk is outlier-driven quality loss; GPTQ (error-compensating) and AWQ (protecting salient channels) make 4-bit practical. I'd always A/B it (compare old and new versions on the same tests) on our task eval."

- *Weight-only vs W8A8?* Weight-only saves memory/bandwidth; W8A8 also uses int8 maths units, useful when compute-bound (large batch, prefill).

### Self-check

1. int4 bytes per parameter with one fp16 scale per 64 weights? → (4 + 16/64)/8 = 0.53 bytes.
2. Why did per-tensor int4 zero out 92% of weights? → Outliers made the step size (0.07) five times larger than a typical weight (0.0135).
3. What does AWQ protect? → Weight channels that see large activations.

### TL;DR

Fewer bits → less memory, often faster decode. Per-group scales, GPTQ/AWQ for 4-bit, FP8/int8 near-lossless; always re-evaluate.

---

## 11.8 Distillation and small specialist models

### What it is

**Knowledge distillation** (Hinton, Vinyals & Dean, 2015) trains a small **student** model to imitate a large **teacher**. Classic distillation matches the teacher's full probability distribution ("soft targets", softened by a **temperature** — a knob that flattens or sharpens probabilities), which carries more information than one correct label; for LLMs teams often just fine-tune the student on **teacher-generated answers** to real product prompts.

#### Why it matters

A student sized for a narrow task serves the high-volume path at a fraction of the cost and latency. **DistilBERT** kept about 97% of BERT's (a 2018 Google language-understanding model) language-understanding score with 40% fewer parameters and ~60% faster inference.

### How it works

1. Collect representative prompts from traffic (with consent and privacy review).
2. Have the teacher answer them; filter bad answers.
3. Fine-tune the student on the pairs (optionally on teacher probabilities).
4. Evaluate on a **held-out golden set** (fixed test cases with known answers, never used for training); deploy behind a router (§11.14) that escalates hard cases.

### Trade-offs and cautions

Students inherit the teacher's mistakes and are good only **inside** the training distribution; retrain when the teacher or product changes. **Check provider terms** — several forbid using outputs to build competing models.

### In the interview

"Distil when a narrow, high-volume task clears the quality bar on a small model: generate teacher outputs on real traffic, fine-tune a student, gate it with evals, and route to it with escalation."

### TL;DR

Teach a small model your task using a big model's answers; serve the small one on the volume path.

## 11.9 Speculative decoding

### What it is

A cheap **draft** model guesses the next *k* tokens; the big **target** model checks all *k* in **one** parallel pass (like prefill). Accepted guesses are kept; at the first rejection the target supplies a corrected token. Because decode is bandwidth-bound, verifying 5 tokens costs about as much as generating 1, so accepted guesses are nearly free.

### How it works — the exact acceptance rule (Leviathan et al. 2023; Chen et al. 2023)

For each drafted token *x* (draft probability q(x), target probability p(x)):

1. Accept with probability **min(1, p(x)/q(x))**.
2. If rejected, sample a replacement from the **residual** distribution ∝ max(0, p − q) (∝ means "proportional to") and stop.
3. If all *k* are accepted, sample one bonus token from p.

This rule provably yields **exactly the target's distribution**, so quality is unchanged (though sampled text still varies run to run). With acceptance probability α = Σ min(p, q), expected tokens per target pass = **(1 − α^(k+1)) / (1 − α)** (assuming each guess is accepted independently).

#### Code

`code/speculative_decoding.py` uses fixed toy distributions over 6 tokens (α = 0.80) and checks both claims empirically.

```
def speculative_step(k):
    drafts = rng.choice(len(p), size=k, p=q)          # draft model runs k cheap steps
    out = []
    for x in drafts:                                  # target scores all k at once
        if rng.random() < min(1.0, p[x] / q[x]):      # accept with prob min(1, p/q)
            out.append(x)
        else:                                         # reject: resample from residual
            resid = np.maximum(p - q, 0)
            out.append(rng.choice(len(p), p=resid / resid.sum()))
            return out
    out.append(rng.choice(len(p), p=p))               # all accepted: bonus token
    return out
```

Output:

```
acceptance rate a = sum(min(p, q)) = 0.80
k=1: tokens per target pass = 1.80 (theory 1.80); max |freq - p| = 0.0009
k=2: tokens per target pass = 2.44 (theory 2.44); max |freq - p| = 0.0017
k=4: tokens per target pass = 3.36 (theory 3.36); max |freq - p| = 0.0009
k=8: tokens per target pass = 4.33 (theory 4.33); max |freq - p| = 0.0012
target p : {'the': 0.3, 'cat': 0.25, 'sat': 0.15, 'on': 0.12, 'mat': 0.1, '.': 0.08}
emitted  : {'the': 0.299, 'cat': 0.25, 'sat': 0.15, 'on': 0.12, 'mat': 0.1, '.': 0.08}
```

Emitted frequencies match the target to three decimals and tokens per pass match the formula. Diminishing returns: k = 4 → 8 adds ~1 token per pass while doubling draft work.

### Trade-offs

- Speed-up depends on **acceptance rate** on your traffic, draft cost and load: published results are often 2–3× on favourable workloads, and gains shrink at large batch because the GPU is already

busy.

- The draft adds memory and complexity. Variants: **Medusa** (extra prediction heads instead of a separate draft), **EAGLE**, and **prompt-lookup / n-gram** (a run of n consecutive tokens) drafting (copy continuations from the prompt — good for edits and RAG answers that quote sources).

### In the interview

> **30-second answer:** "A small draft proposes several tokens; the target verifies them in one parallel pass with an accept/reject rule that preserves its exact output distribution. Decode steps become multi-token steps; gains depend on acceptance rate and load, so benchmark it."

### Self-check

1. With α = 0.6 and k = 3, expected tokens per pass? → $(1 - 0.6^4)/(0.4) = (1 - 0.1296)/0.4 \approx 2.18$.
2. Does speculative decoding change quality? → Not with the exact acceptance rule.

### TL;DR

Guess cheaply, verify in bulk: same distribution, fewer expensive passes.

---

## 11.10 GPU memory sizing

### What it is

Check that the model **fits** and how many users it serves at once:

```
GPU memory needed ≈ weights + KV cache + activations/runtime overhead
weights           ≈ parameters × bytes per parameter   (fp16/bf16 = 2, int8/fp8 = 1, int4 ≈ 0.5 +
scales)
KV cache          ≈ concurrent sequences × tokens per sequence × KV bytes per token   (§11.4)
overhead          ≈ activations, CUDA context, allocator slack — engines typically reserve a fraction
```

**Mixture-of-experts (MoE)** models (many "expert" sub-networks, with a small router choosing a few per token) need **all** experts in memory though each token uses a few: Mixtral 8×7B has ~47B parameters but uses ~13B per token — memory follows the 47B, compute the 13B.

### How it works — worked examples

`code/serving_calculator.py` implements these formulas, assuming the engine may use 90% of GPU memory (vLLM's default `gpu_memory_utilization`) with activations in the other 10% — a simplification.

```python
def kv_bytes_per_token(m, kv_bytes=2):
    return 2 * m.layers * m.kv_heads * m.head_dim * kv_bytes

def fit(m, n_gpus, gpu_gb, bytes_per_param, ctx_len, util=0.90):
    usable = n_gpus * gpu_gb * 1e9 * util
    free = usable - m.params * bytes_per_param        # what is left after the weights
    tokens = free / kv_bytes_per_token(m)
    return tokens, int(tokens // ctx_len)              # KV room, concurrent sequences
```

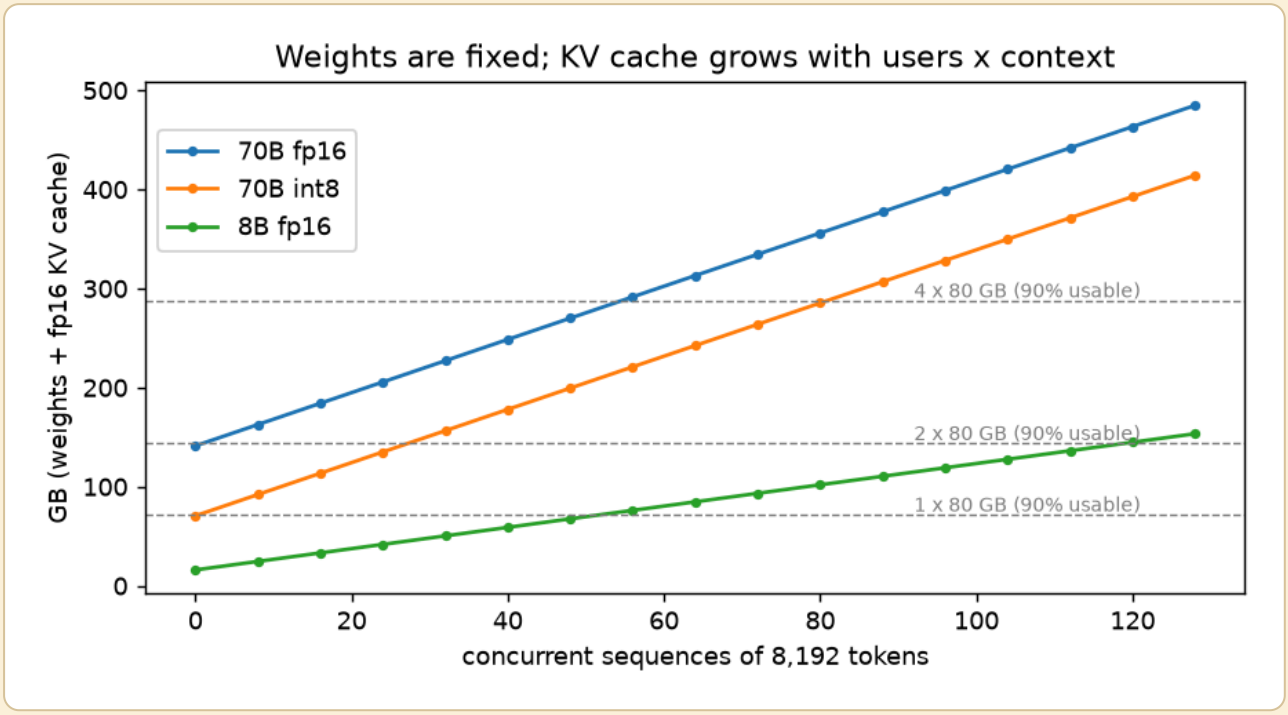
Output (section 2 of the script):

```
Llama-3-8B (GQA)     fp16 on 1 x 24 GB    weights   16.0 GB | KV room    42,725 tokens =   10
concurrent 4K sequences
Llama-3-8B (GQA)     fp16 on 1 x 80 GB    weights   16.0 GB | KV room   427,246 tokens =  104
concurrent 4K sequences
Llama-3-70B (GQA)    fp16 on 2 x 80 GB    weights  141.2 GB | KV room     8,545 tokens =    2
concurrent 4K sequences
Llama-3-70B (GQA)    fp16 on 4 x 80 GB    weights  141.2 GB | KV room   447,998 tokens =  109
concurrent 4K sequences
Llama-3-70B (GQA)    int8 on 2 x 80 GB    weights   70.6 GB | KV room   223,999 tokens =   54
concurrent 4K sequences
Llama-3-70B (GQA)    int4 on 1 x 80 GB    weights   35.3 GB | KV room   112,000 tokens =   27
concurrent 4K sequences
```

Step through one row by hand — **70B at fp16 on two 80 GB GPUs**:

1. Usable memory = 2 × 80 × 0.9 = 144 GB.
2. Weights = 70.6 × $10^9$ × 2 B = 141.2 GB. Left for KV: 2.8 GB.
3. KV per token = 320 KiB = 327,680 B → 2.8 GB / 327,680 B ≈ 8,500 tokens → **two** 4K conversations. It "fits" but is useless as a service.
4. With 4 GPUs, 288 − 141.2 ≈ 147 GB is left → ~448,000 tokens → **109** sequences. int8 on 2 GPUs gives **54**.

So "does 70B fit?" has honest answers: barely on 2×80 GB at fp16, comfortably on 4×80 GB, or on 2×80 GB at int8 (1×80 GB at int4, before the ~3% for group scales) — each a quality/cost trade-off.




The script's plot shows memory versus concurrent 8K-token sequences: weights set the intercept, KV cache the slope; where a line crosses a dashed "N × 80 GB" line is that setup's concurrency limit.

## Common mistakes

- Forgetting the KV cache (the most common sizing error).
- Using parameter count × 4 bytes (fp32) — serving is almost always 16-bit or lower.
- Assuming fit implies acceptable speed; mixing GB ($10^9$) and GiB ($2^{30}$), a 7% difference.

### In the interview

> **30-second answer:** "Weights are parameters × bytes: 70B at bf16 is ~140 GB. KV cache is 2 × layers × KV-heads × head-dim × bytes per token — ~320 KiB for Llama-3-70B — times users times context, plus ~10% overhead. On 2×80 GB the weights leave no cache room, so 4 GPUs with tensor parallelism, or int8/FP8 on 2."

### Self-check

1. Raw weights of a 13B model at bf16? → ~26 GB.
2. Why can a model fit yet be unusable? → No KV room, so almost no concurrency.
3. MoE memory — total or active parameters? → Total.

### TL;DR

Memory = params × bytes + users × context × KV-per-token + overhead. Size for concurrency, not "it loads".

---

## 11.11 Model parallelism: tensor and pipeline

### What it is

When one GPU cannot hold or serve a model fast enough, split the work:

| Kind | What is split | Communication | Typical use |
|---|---|---|---|
| **Replicas (data parallelism)** | Nothing — each GPU group has a full copy; requests are load-balanced (spread evenly by a load balancer — see System Design Fundamentals) | None | Scale **throughput** once one replica fits |
| **Tensor parallelism (TP)** | Each layer's weight matrices are sliced across GPUs (Megatron-LM style); every GPU computes part of every layer | An **all-reduce** (every GPU sums its partial results with all the others) **in every layer** → needs very fast GPU-to-GPU links such as NVIDIA's **NVLink** | Within one node (one server machine, usually up to 8 GPUs); also lowers per-token latency |
| **Pipeline parallelism (PP)** | Consecutive **layers** go to different GPUs (stage 1 = layers 1–40, stage 2 = 41–80) | Activations passed between stages once per step | Across nodes, when the model is too big for one node |
| **Expert parallelism (EP)** | Different MoE experts on different GPUs | Tokens routed to the GPU holding their expert | Large MoE models |

Analogy: TP is four cooks chopping one onion together (fast, constant coordination); PP is an assembly line passing a tray between stations.

### How it works

- **TP:** in `Y = X·W` with `W` split by columns across 4 GPUs, each GPU computes a quarter of `Y`; the next matrix is split by rows, so its partial sums are all-reduced. That is two communications per transformer layer (one after attention, one after the MLP) — fine over NVLink, slow over ordinary networking.
- **PP:** with one request in flight, stage 2 idles while stage 1 works (a **pipeline bubble**); engines keep several **micro-batches** (small slices of the batch) in flight so all stages stay busy.

In vLLM these are single settings: `--tensor-parallel-size 4` (and `--pipeline-parallel-size` for multi-node).

### Trade-offs and common mistakes

TP adds communication, so 8-way TP of a small model can be *slower* than one GPU — use replicas. PP adds capacity, not single-request speed. Mistake: spreading TP across nodes over slow networking.

### In the interview

"Replicas scale throughput; TP splits each layer across GPUs in a node over NVLink; PP splits layers into stages across nodes, with micro-batching to avoid bubbles. For 70B: TP=4 in a node, then replicas behind a load balancer."

### TL;DR

Fit with TP (inside a node) or PP (across nodes); scale throughput with replicas.

---

## 11.12 Streaming responses

### What it is

**Streaming** sends tokens to the client as they are generated instead of waiting for the full answer. LLM APIs typically use **Server-Sent Events (SSE)** — a long-lived **HTTP** (the web's request–response protocol) response carrying a sequence of small events (Anthropic's API emits `message_start`, `content_block_delta` events carrying text, and `message_stop`; OpenAI's Chat Completions emits `chat.completion.chunk` objects).

### Why it matters

Streaming does **not** cut compute or cost, but an answer that starts after 0.5 s feels responsive. It also enables **early cancellation**: if Stop reaches the engine, the remaining decode work (and billed output tokens) is saved; merely hiding output in the UI saves nothing.

### How it works — client code (needs an API key — not run)

```
import anthropic

client = anthropic.Anthropic()                    # reads ANTHROPIC_API_KEY
with client.messages.stream(
    model="claude-opus-5-5",
    max_tokens=4096,
    messages=[{"role": "user", "content": "Explain the KV cache in two sentences."}],
) as stream:
    for text in stream.text_stream:               # yields text as it is generated
        print(text, end="", flush=True)
    final = stream.get_final_message()            # full message + usage at the end
print("\nusage:", final.usage.input_tokens, "in /", final.usage.output_tokens, "out")
```

A server relaying this to browsers re-emits SSE (or uses **WebSockets**, an always-open two-way browser connection) and records TTFT when the first chunk is flushed.

### Trade-offs and common mistakes

- **Safety vs speed:** output checks cannot see the whole answer before it is shown. Buffer a sentence or a few hundred characters, run a streaming classifier, and keep a **stop control** — which cannot un-show displayed text.

- **Structured output:** streamed JSON is invalid until complete; parse incrementally or wait.
- Buffering **proxies** (servers in the middle that forward traffic) silently break streaming; long streams need timeouts and **heartbeats** (small periodic "still alive" messages).

### In the interview

"Streaming improves perceived latency and lets cancellation stop decode early, but doesn't cut cost by itself. Stream over SSE, buffer enough for output checks, make sure cancel reaches the engine."

### TL;DR

Stream for perceived speed; buffer for checks; cancellation must reach the GPU to save money.

---

## 11.13 Caching: exact, semantic and prefix

Three caches at three layers — keep them apart:

| Cache | Key | What it saves | Where it lives |
|---|---|---|---|
| **Exact response** | Normalized request (+ tenant — the customer organisation — model, settings) | The whole call | In front of the model (e.g. Redis, a fast in-memory key–value store) |
| **Semantic response** | Embedding similarity ≥ threshold | The whole call, for paraphrases | In front (vector index) |
| **Prefix / prompt** | Exact token prefix | Prefill of the shared part only | Inside the engine or provider |

### What it is

- **Exact cache:** same question, same context → stored answer. Low **hit rate** (share of requests answered from the cache) for free text, high for requests like "summarize document v17".
- **Semantic cache:** embed the request (an **embedding** is a vector capturing meaning — Chapter 3), search earlier requests, and reuse an answer if cosine similarity ≥ a threshold. Catches paraphrases ("reset password" vs "change my pwd").
- **Prefix (prompt) cache:** the engine keeps the KV cache of a prompt's beginning (**system prompt** — the standing instructions sent before the conversation — tools, a shared document) and reuses it when a request starts with **exactly** the same tokens. Hosted APIs sell this as discounted **cached input** — Anthropic bills cache reads at 10% of the base input price on most models, less on some (5% on Claude Opus 5.5), with a 5-minute cache write at 1.25×; OpenAI caches long repeated prefixes automatically. Minimum lengths and lifetimes vary by provider.

#### Why it matters

Support bots see many repeated questions, and a cached answer costs almost nothing. Prefix caching cuts cost and TTFT for long shared prompts.

### How it works — the design rules

1. **Stable content first:** tools → system prompt → shared documents → conversation → new question. One changing token early (a timestamp, a user ID) breaks every prefix hit after it.
2. **Scope every response-cache entry** by tenant, permissions, model, prompt and source-document versions — never serve one customer's answer to another.
3. **Exclude or key** personalized and time-sensitive answers ("my order status").
4. **Expire** entries with a TTL (time to live) and invalidate them when documents change.
5. **Tune the similarity threshold on labelled pairs** of same/different-meaning questions.

6. For personalized flows, cache the **retrieved context** instead of the answer.

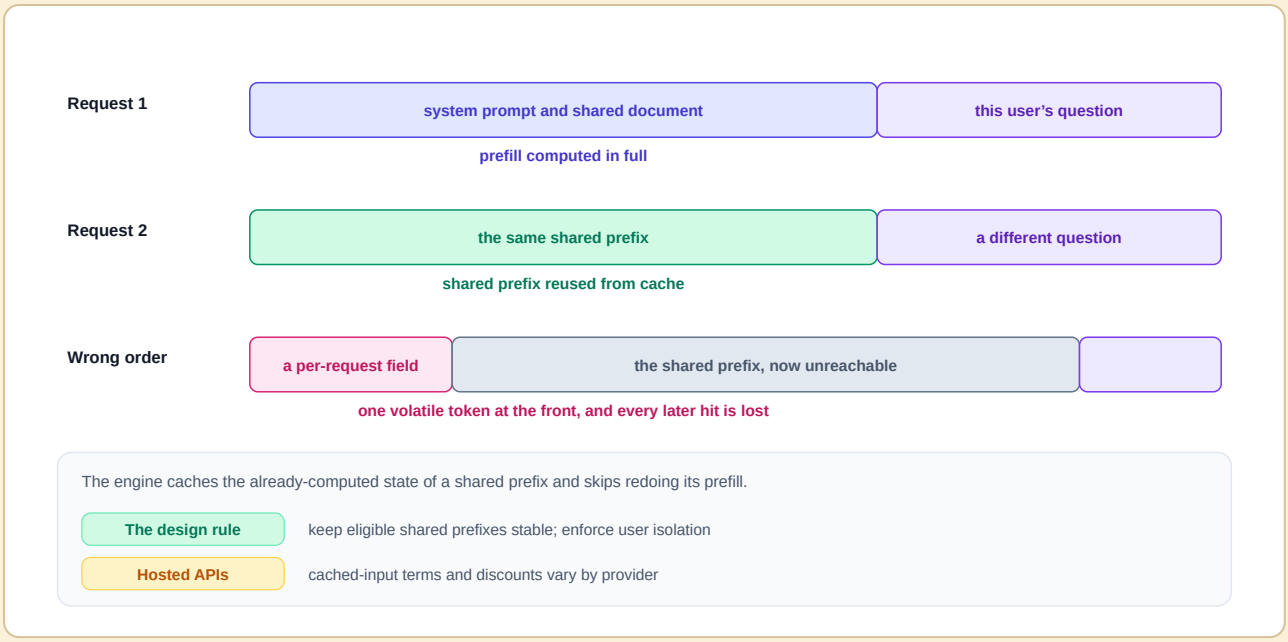



*Figure — Prefix caching only works if the shared part comes first (course diagram).*

Prefix caching with the Anthropic SDK (needs an API key — not run):

```
response = client.messages.create(
    model="claude-opus-5-5",
    max_tokens=4096,
    system=[{
        "type": "text",
        "text": LONG_STATIC_POLICY_DOCUMENT,              # identical on every request
        "cache_control": {"type": "ephemeral"},          # cache everything up to here
    }],
    messages=[{"role": "user", "content": user_question}],  # volatile part goes last
)
u = response.usage
print(u.cache_creation_input_tokens, u.cache_read_input_tokens, u.input_tokens)
```

If `cache_read_input_tokens` stays at zero across repeated calls, something in the prefix is changing — or the prefix is shorter than the model's minimum cacheable length.

### Code — exact + semantic cache with a local fake embedder

`code/semantic_cache.py` uses a deterministic stand-in embedder (synonym normalization + hashed words and character **trigrams**, i.e. three-letter chunks) so it runs offline.

```
def get(self, tenant, text, now):
    k = (tenant, self._norm(text))
    if k in self.exact and now - self.exact[k][1] < self.ttl:     # layer 1: exact
        return self.exact[k][0], "exact", 1.0
    v = self.emb.embed(text)
    best, best_sim = None, -1.0
    for t, ev, ans, ts, orig in self.entries:                       # layer 2: semantic
        if t != tenant or now - ts >= self.ttl:                     # scope + freshness first
            continue
        sim = float(v @ ev)                                         # cosine (unit vectors)
        if sim > best_sim:
            best, best_sim = (ans, orig), sim
    if best and best_sim >= self.threshold:
        return best[0], f"semantic (matched '{best[1]}')", best_sim
    return None, "miss", best_sim
```

Output:

```
[acme] 'How do I reset my password?'                -> miss
sim=-1.00
[acme] 'how do i reset my password'                 -> exact
sim=1.00
[acme] 'How can I change my pwd?'                   -> semantic (matched 'How do I reset my
password?') sim=1.00
[acme] 'Please tell me how to modify my password'   -> semantic (matched 'How do I reset my
password?') sim=1.00
[acme] 'how do I reset my pasword'                  -> miss
sim=0.67
[acme] 'How do I cancel my subscription?'           -> miss
sim=0.05
[acme] 'How do I terminate my sub?'                 -> semantic (matched 'How do I cancel my
subscription?') sim=1.00
[acme] 'How do I cancel my order?'                  -> miss
sim=0.50
[globex] 'How do I reset my password?'                -> miss
sim=-1.00
[acme] 'Can I get my money back?'                   -> miss
sim=0.14

LLM calls: 6 of 10 requests
similarity 'cancel subscription' vs 'cancel order' = 0.50 (threshold 0.85 keeps them apart; 0.40 would
wrongly merge them)
same question 2 h later: miss (TTL expired)
```

What it shows: exact hits after normalization; semantic hits for paraphrases; a **near-miss** ("cancel my order" vs "cancel my subscription", 0.50) rejected at threshold 0.85 but wrongly merged at a loose one; a typo missing at 0.67 (real embedders handle typos better, but the threshold trade-off is the same); **tenant isolation** (globex never sees acme's entries); and **TTL expiry**. The perfect 1.00 scores are an artefact of the toy synonym table; real embedding models give values like 0.85–0.95 for paraphrases.

### Trade-offs

Semantic caching trades a small risk of a wrong answer for savings; the cost of a wrong answer (a refund policy vs a joke) sets how strict to be. Measure **hit rate** and **wrong-hit rate** on real traffic before promising savings; the lookup itself costs a little.

### In the interview

"Three layers: an exact cache keyed by normalized request plus tenant and versions; a semantic cache with a threshold tuned on labelled pairs, TTLs and invalidation; and prefix caching in the engine or provider, which needs static content first. The main risk is serving a near-miss or another user's answer, so scoping comes before hit rate."

### Self-check

1. Does prefix caching skip output generation? → No, only the prefill of the shared prefix.
2. Why put the timestamp last? → A change at the start invalidates the whole cached prefix.
3. Two metrics for a semantic cache? → Hit rate and wrong-hit rate.

### TL;DR

Exact cache for repeats, semantic for paraphrases (thresholded, scoped), prefix for shared prompt beginnings.

---

## 11.14 Routing and cascades to cheaper models

### What it is

A **router** decides up front which model answers; a **cascade** tries a cheap model first and escalates when a check fails.

### How it works

Router signals, from cheapest to most expensive:

1. **Heuristics** — task type (summaries, classification, drafts → small model), prompt length, keywords.
2. **A trained classifier** — a small model trained on labelled outcomes ("did the small model's answer pass?").
3. **A verifier** — the cheap model answers, then a check (schema validation, citation check, grader model) decides whether to escalate. A model's self-reported confidence is often poorly **calibrated** (its stated confidence does not match how often it is right); test it against real outcomes first.

Savings: if 75% of traffic can use a model at 1/10 the price, cost × (0.75 × 0.1 + 0.25 × 1) = **0.325** (67.5% saved), assuming equal token counts. A cascade also pays for the cheap attempt on escalations.

#### Code — router vs cascade

`code/router_and_limits.py` runs 8 tasks of varying difficulty through two fake models (small: $0.3/$1.2 per M tokens, handles difficulty ≤ 2; large: $3/$12, handles everything).

```python
def rule_router(task):                      # decide first, before any model runs
    easy_kinds = {"summarize_thread", "status_draft", "classify"}
    return SMALL if task["kind"] in easy_kinds and task["tin"] < 4000 else LARGE

for t in TASKS:                             # cascade: small first, escalate on failure
    ans, tin, tout = SMALL.run(t)
    c = SMALL.cost(tin, tout)
    if not verifier(ans):
        escalations += 1
        ans, tin, tout = LARGE.run(t)
        c += LARGE.cost(tin, tout)
```

Output:

```
all on large model : $0.08796 for 8 tasks, 0 bad answers
rule router        : $0.05038 (43% cheaper), 1 bad answer(s) slipped through
cascade + verifier : $0.03400 (61% cheaper), escalation rate 2/8, 0 bad answers
cost telemetry by (model, feature):
  large answer_question   $0.01380
  large summarize_thread  $0.01140
  small answer_question   $0.00240
  small classify          $0.00025
  small status_draft      $0.00168
  small summarize_thread  $0.00447
```

The rule router let one hard "summarize" task through to the small model (a quality miss) and sent an easy question to the large one (a cost miss) — routing's two failure modes. The cascade won here because its verifier was perfect; real verifiers miss things and add latency, and escalations pay twice.

## Trade-offs

| | Router | Cascade |
|---|---|---|
| Latency | One call | Two calls on escalation |
| Needs | Good up-front signal | A reliable verifier |
| Failure mode | Misroutes silently | Pays twice for hard requests |

## Common mistakes

- Routing by task name without measuring quality per route; not watching the **escalation rate**.
- Forgetting that prompt caches are per model, so splitting traffic lowers hit rates.

## In the interview

"Most requests are simple, so a cheap model serves them and the strongest model handles the rest — via a router that decides up front or a cascade that escalates on a failed check. I'd validate the split on a golden set and monitor escalation rate and sampled quality per route."

## TL;DR

Most traffic is easy: cheap model first, escalate on failure, measure cost and quality per route.

---

## 11.15 Cost estimation: back-of-the-envelope for AI systems

### What it is

Rough, out-loud arithmetic turning "users and features" into tokens, dollars and GPUs. It tells you which optimization matters.

**The anchors (illustrative — verify current prices)**

| Anchor | Value used in this chapter |
|---|---|
| Token size | ~4 characters ≈ 0.75 words |
| Large hosted model | $3 per M input, $12 per M output |
| Small hosted model | ~10× cheaper ($0.3 / $1.2) |
| One chat turn | a few thousand tokens in, a few hundred out ≈ 1 cent on the large model |
| Self-hosted node | $3–4/hour, sustaining 2,500 output tokens/s on *your* workload (must be load-tested) |
| Raw weights | 2 bytes per parameter in fp16/bf16 |
| Raw vectors | chunks × dimensions × 4 bytes (float32) |

### How it works — the unit-economics habit

1. **Size one user-visible unit** (a turn, a conversation, a document) in tokens.
2. **Price the unit.**
3. **Multiply by volume**, then state the conclusion — which optimization you would try first.

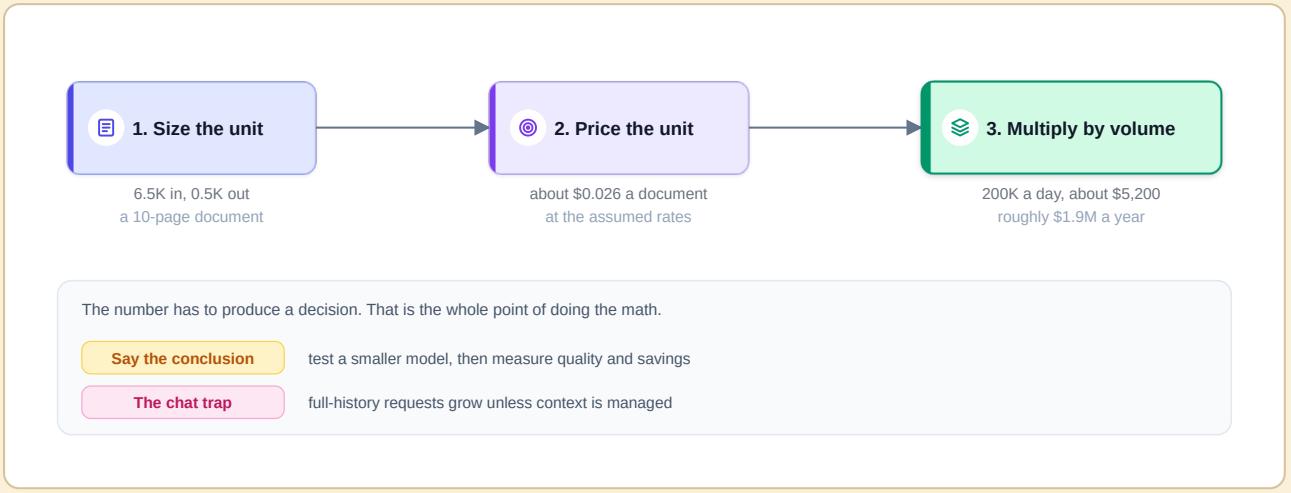



*Figure — Size, price, multiply, decide (course diagram).*

**Example A — one turn:** 3,000 in, 300 out → 3,000 × $3/M + 300 × $12/M = $0.009 + $0.0036 = **$0.0126**.

**Example B — support bot:** 100,000 conversations/day × 10 turns, averaging 3K in / 300 out per turn → 30K in + 3K out per conversation = **$0.126 → $12,600/day ≈ $378,000 per 30 days**, before retrieval, retries and monitoring. Conclusion: summarize history, route easy turns, cache.

**Example C — document summarization:** a 10-page document ≈ 5,000 words ≈ 6,700 tokens in (5,000 ÷ 0.75); a 400-word summary ≈ 500 tokens out → $0.0201 + $0.006 ≈ **$0.026 per document**; 200K documents/day ≈ **$5,220/day ≈ $1.9M/year**. Conclusion: test a small model on representative documents — length alone does not make a task hard.

**The chat-history trap:** if every turn re-sends the whole conversation, input grows each turn. With a 500-token system prompt and 200-token user and assistant messages, turn $i$ sends $500 + 200 \times (2i - 1)$ tokens, so 10 turns send **25,000** input tokens instead of 7,000 (10 × 700) — 3.6× more, and the total

grows quadratically with conversation length. Fixes: summarize or window old turns, prefix-cache the stable part.

**Self-host fleet:** peak 100 requests/s × 400 output tokens = 40,000 tokens/s. At a measured 2,500 tokens/s per node → 16 nodes, + **headroom** (spare capacity for spikes and failures) → 20. At $3/node-hour → 20 × 3 × 24 × 365 = **$525,600/year** before networking, storage and people (benchmark with your real input lengths).

**Vector storage for RAG:** 10M chunks × 768 dims × 4 B = **30.72 GB** of raw vectors (before index overhead, metadata, text and replicas); 1B chunks ≈ **3.07 TB** → a **sharded** design (data split across many machines; see System Design Fundamentals).

### API vs self-hosting

| Factor | Favours a hosted API | Favours self-hosting |
|---|---|---|
| Volume | Low or spiky | High and steady |
| Quality needed | Only frontier models are good enough | A small or open model passes your evals |
| Data governance | Provider terms suffice | Requirements hosted options can't meet |
| Team | No inference team | Infra team, strategic priority |
| Control | Rate limits and deprecations (the provider retiring a model) acceptable | Need control of versions, capacity, latency |

Compute **break-even** (the volume where both options cost the same) on total cost (people, **redundancy** — spare machines in case one fails — and **on-call**, engineers on a rota to fix incidents at any hour), not GPU price alone. The usual end state is a **mix**: frontier API for hard, low-volume paths, a cheap small model for the easy bulk, a router in between.

## Common mistakes

- A total with no per-unit number; assumptions stated as facts.
- A conclusion that flips if one assumption doubles — find and confirm that assumption first.

## In the interview

> **30-second answer:** "Let me size one unit in tokens, price it and multiply by volume, with prices to confirm against today's list. The per-unit and per-day numbers point at the next step: route the simple bulk to a cheaper model, cache repeats, shorten history. With large, stable volume I'd also cost a GPU fleet — peak output tokens/s over measured per-node throughput, plus headroom — and compare it with the API bill at equal quality."

## Self-check

1. 1,000 words in, 200 words out at $3/$12 per M? → ~1,333 × 3/M + ~267 × 12/M ≈ $0.004 + $0.0032 ≈ $0.0072.
2. Why is a 10-turn chat not 10× one turn? → History is re-sent every turn.
3. Compare a fleet estimate against? → The API bill at equal quality and availability.

## TL;DR

Per unit → price → volume → decision. Say assumptions aloud; make the conclusion survive a 2× error.

---

## 11.16 Rate limiting, backpressure and provider risk

### What it is

Hosted APIs limit **requests per minute** and **tokens per minute** (Anthropic counts input and output tokens separately) and return HTTP **429** (the "too many requests" status code) when you exceed them. Your service should also limit **its** users so one tenant cannot consume the whole budget.

### How it works

- **Token bucket:** a bucket of capacity $C$ tokens refills at $C/60$ per second; a request of $n$ tokens proceeds only if $n$ are available — bursts allowed, average enforced.
- **Backpressure:** when empty, queue (bounded), return "busy", or **degrade** (smaller model, shorter context). Retry 429s with **exponential backoff** (wait 1 s, 2 s, 4 s …) plus random **jitter** (a little random extra delay so clients don't all retry at once), honouring `retry-after`.
- **Budgets:** per-org and per-feature spend limits with alerts, and a **kill switch** per feature.
- **Outages:** a **degradation ladder** — canned answers for common intents → second provider or self-hosted small model → an honest "temporarily unavailable".

```
class TokenBucket:
    def __init__(self, tokens_per_min):
        self.capacity = self.level = tokens_per_min
        self.rate, self.t = tokens_per_min / 60.0, 0.0
    def try_take(self, n, now):
        self.level = min(self.capacity, self.level + (now - self.t) * self.rate)
        self.t = now
        if n <= self.level:
            self.level -= n
            return True
        return False
```

Output (end of `router_and_limits.py`):

```
token bucket 20K tok/min, 12 x 2.8K-token requests in 3 s: 7 admitted, 5 queued/degraded
```

The bucket absorbed 7 requests (19.6K tokens; 3 s of refill adds only 1K) and pushed back the rest instead of triggering 429s.

### In the interview

"Treat the provider like any dependency: client-side token budgets, bounded queues, jittered retries, a fallback model or provider, per-tenant quotas, kill switches and a defined degraded mode."

### TL;DR

Budget tokens before the provider does; degrade gracefully.

---

## 11.17 Observability

### What it is

**Observability** means answering "what is happening and why?" from **metrics** (numbers over time), **traces** (one request's path through gateway → cache → router → model → checks) and **logs**.

### What to measure

| Area | Metrics |
|---|---|
| Latency | TTFT, TPOT, E2E at p50/p95/p99; queue time |
| Capacity | Requests/s, tokens/s, GPU utilization, **KV-cache usage %**, pre-emptions |
| Cost | Tokens per request; **cost per request, feature, tenant, model** |
| Efficiency | Cache hit and wrong-hit rates, prefix-cache hit rate, router split, escalation rate |
| Reliability | Error and 429 rates, timeouts, fallback activations |
| Quality & safety | Sampled grader scores per path, golden-set results per change, refusal and guardrail-block rates, user feedback |

#### How it works

1. Emit metrics from the engine (vLLM exposes **Prometheus** — a popular open-source metrics system — metrics such as KV-cache usage and running/waiting requests) and from your gateway.
2. Trace each request with IDs linking prompt version, model, route, cache result and token counts (**OpenTelemetry**, an open standard for traces and metrics, has GenAI semantic conventions; LLM-specific tools include Langfuse, LangSmith and Arize Phoenix).
3. Put **cost on the same dashboard as latency** and alert on **drift** (a slow, unplanned change in a metric) — a prompt edit that doubles input size is a production regression.
4. Run evaluations continuously on the cheap paths (router, cache, distilled model), not just at launch (Chapter 8).
5. Prompts and outputs may contain personal data — redact, restrict access, set **retention** (how long data is kept) (Chapter 12).

#### In the interview

"TTFT/TPOT percentiles, throughput and KV-cache pressure for capacity; tokens and dollars per request by feature, tenant and model; cache-hit and escalation rates; sampled quality scores on every cheap path — with alerts on cost drift as well as latency."

#### TL;DR

Latency, capacity, cost and quality on one dashboard, with alerts on all four.

---

## 11.18 Capstone: pricing the inference layer of a 5M-user product

**The scenario:** a project-management **SaaS** (software as a service — software rented over the internet) with 5 million daily active users (DAU) plans an in-app AI assistant that explains project status, drafts updates and condenses long threads. You own the model-serving layer: propose a design and price it.

**Assumptions (state them, invite correction):** 20% of DAU use it daily, 4 interactions each; with retrieved context, 2,500 input and 300 output tokens on average; peak = 3× average; prices as in §11.15.

### Step 1 — volumes and the naive bill

- Interactions: 5M × 20% × 4 = **4M/day** → 46.3/s average, ×3 ≈ 139/s peak (plan for ~150).
- Tokens: 4M × 2,500 = **10B input/day**; 4M × 300 = **1.2B output/day**.
- Cost on the large model: 10B × $3/M = $30,000 + 1.2B × $12/M = $14,400 → **$44,400/day ≈ $16.2M/year**.

- Per active user-day: $44,400 / 1M = **$0.044**; a daily user for 30 days costs ≈ $1.33 — about **6.7% of a $20/month seat**. A real budget item, not automatically a margin-killer.

Always produce this naive number first — it justifies every optimization that follows.

### Step 2 — the optimized stack, priced step by step

1. **Route** 75% (summaries, drafts — validated on a golden set) to a small model at 1/10 price: × 0.325 → **$14,430/day**.
2. **Prefix caching:** assume 50% of input tokens are cache hits at a 90% discount → input charges fall by 45% (0.5 × 0.9); output unchanged: input $9,750 → $5,362.50; plus output $4,680 → **$10,042.50/day**.
3. **Result cache** (exact, keyed by thread version, tenant and permissions): assume 15% of requests skip the model → × 0.85 → **$8,536/day ≈ $3.1M/year**, **80.8% below naive** before lookup and eval costs.
4. **Sensitivity:** at 25% cache hits ≈ $7,532/day. Say which assumption you would verify first.

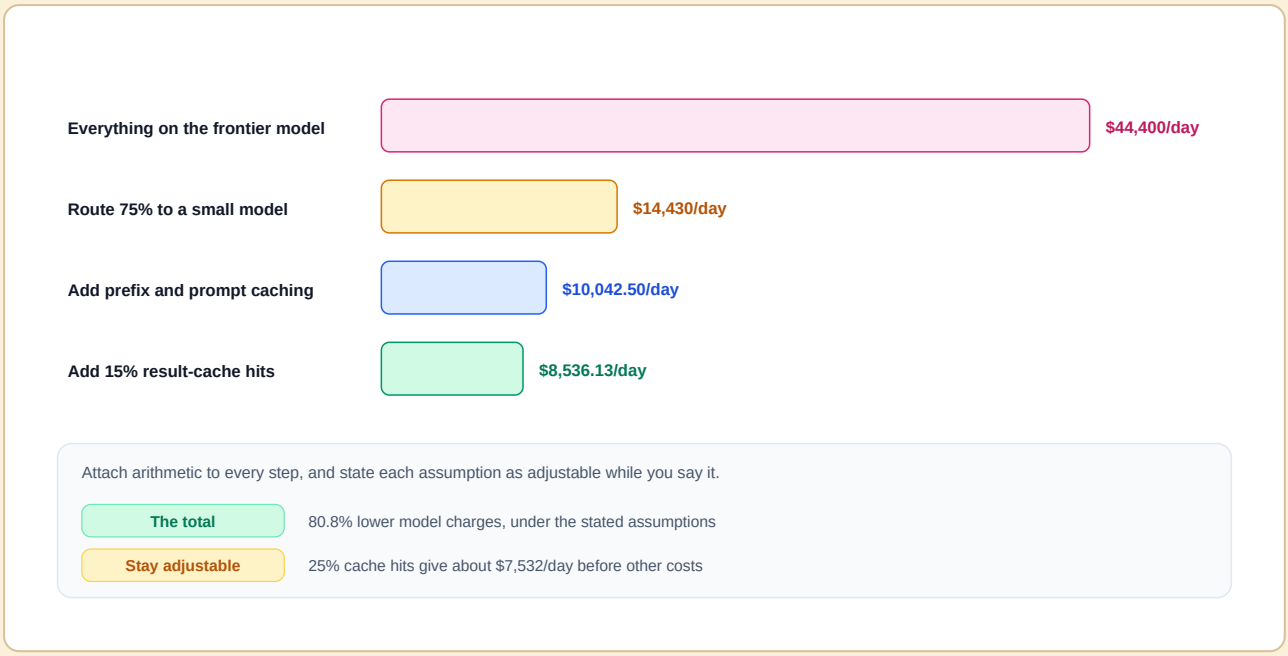



*Figure — From $44,400/day to about $8,536/day under the stated assumptions (course diagram).*

### Step 3 — should the small-model path be self-hosted?

- Calls after caching: 4M × 75% × 85% = **2.55M/day**; × 300 output = 765M output tokens/day = 8,854/s average, ×3 ≈ **26,562/s peak**.
- At a measured 2,500 tokens/s per node: 10.6 → **11 nodes**, provision **14** for failures and bursts.
- Fleet: 14 × $3 × 24 = **$1,008/day**.
- Same path on the hosted small model: input 6.375B × $0.30/M × 0.55 (prefix caching) = $1,051.88 + output 765M × $1.20/M = $918 → **$1,969.88/day**.
- Difference ≈ $962/day ≈ **$351K/year** *before* engineering, on-call and **failover** (switching to a backup when something breaks) — worth it only if a team exists and quality holds; keep API overflow for peaks.

Section 8 of `serving_calculator.py` reproduces every capstone number; change an assumption and re-run.

### Step 4 — guard the bill and the quality

- **Cost telemetry** per interaction, by feature, model and team, next to latency; alert on drift.
- **Quality on every cheap path:** golden set + sampled grading; watch escalation rate both ways (too high → cost, too low → quality).

- **Budgets and backpressure:** per-org token/spend limits, a tested outage policy, per-feature kill switches.

### Follow-ups worth rehearsing

- *"The prefix-cache hit rate halved overnight."* Check recent **deploys** (releases of new code): someone probably put a per-request field (timestamp, user name) near the top of the prompt. Fix the structure and add a test that asserts cache hits.
- *"Product wants proactive suggestions on every page load."* Re-run the naive math with page-load volume (far higher than explicit requests); that usually forces precomputation, aggressive caching, the smallest viable model — or not shipping it.
- *"Finance wants a fixed cost ceiling per user."* Only with enforced per-org token budgets and a degradation ladder (smaller model → shorter context → polite refusal).
- **Scale changes the answer:** at 50K DAU just use an API; at 50M DAU self-hosting and the router become central.

**What separates levels:** mid-level names caching and a cheaper model; senior produces the naive number first, prices each optimization with an explicit assumption and runs a real break-even; staff ties cost per user to revenue per user and says who owns cost regressions.

---

## Chapter TL;DR

- **Two phases:** prefill (parallel, compute-bound, TTFT) and decode (sequential, bandwidth-bound, tokens/s). E2E ≈ TTFT + TPOT × (n − 1).
- **KV cache:** 2 × layers × KV-heads × head-dim × bytes per token; usually the real concurrency limit; GQA and 8-bit KV shrink it.
- **Batching:** continuous (per-step) batching is standard; **PagedAttention** allocates KV in blocks.
- **Cheaper model:** quantization (int8/FP8 near-lossless; int4 with GPTQ/AWQ), distillation, speculative decoding.
- **Memory:** params × bytes + users × context × KV/token + overhead; TP in a node, PP across nodes, replicas for throughput.
- **Avoid calls:** exact, semantic and prefix caching; route or cascade to cheaper models with evals watching.
- **Cost:** per unit → price → volume → decision; compare API vs fleet on total cost.
- **Operate:** safe streaming, token budgets, backpressure, fallbacks, kill switches; one dashboard for latency, capacity, cost and quality.

## Further reading


- Kwon et al., *PagedAttention* (SOSP 2023, the vLLM paper); Yu et al., *Orca* (OSDI 2022).
- Leviathan et al. (ICML 2023) and Chen et al. (2023) on speculative decoding.
- Frantar et al., *GPTQ* (2022); Lin et al., *AWQ* (2023); Dettmers et al., *LLM.int8()* (2022); Xiao et al., *SmoothQuant* (2022).
- Hinton et al., *Distilling the Knowledge in a Neural Network* (2015); Shoeybi et al., *Megatron-LM* (2019).
- Chip Huyen, *AI Engineering* (O'Reilly, 2025), inference-optimization chapter; docs.vllm.ai; Anthropic and OpenAI API docs.